AETHERIA AI Data-Intelligence Pipeline

Distributed Architecture, Deterministic Quality Gates, Multi-Tenant Partitioning & Scale Strategy

1. Executive Overview & Ingestion Philosophy

AETHERIA is a high-throughput, enterprise-grade multi-vertical intelligence pipeline harvesting live structured data across **Startups, Products, Research Papers, Jobs, and News**. The platform enforces a strict **Deterministic-First Quality Gate**: whenever structured feeds, verified seed catalogs, or official REST APIs provide complete payloads, the engine completely bypasses LLM inference (0 token cost, sub-second execution latency). Unstructured or partial payloads are routed through a resilient multi-tier LLM fallback router with strict Pydantic v2 schema enforcement and 1-shot self-healing validation.

2. Concurrency Baseline vs. 500k+ Production Scale Architecture

Single-Node Demo Baseline (Current Implementation): Concurrency is governed by compartmentalized asynchronous pools: asyncio.Semaphore(15) for HTTP network crawlers with 4s bounded timeouts and domain circuit breakers, and asyncio.Semaphore(3) for LLM router inference. Ingested payloads stream to user-scoped JSONL partitions under data/raw/{user_id}/.
500K+ Horizontal Production Scale-Out: Workers are completely stateless. Scaling to 500,000+ records requires zero structural refactoring: ingestion scales horizontally across worker pools managed by **Celery / Arq** over **Redis** task queues sharded by source domain. High-volume PostgreSQL persistence utilizes **PgBouncer connection pooling**, partitioned tables, and **pgvector / Qdrant** for approximate nearest neighbor semantic deduplication.

CURRENT IMPLEMENTED ARCHITECTURE
- 5 Parallel Harvesters (arXiv, YC, HF Spaces, RSS, APIs)
- Resilient Async HTTP Client with Domain Circuit Breakers
- Hybrid Structuring Gate (Deterministic Bypass > 90%)
- RapidFuzz Deterministic Entity Canonicalization Engine
- Supabase PostgreSQL Persistence with Row-Level Security
- User-Scoped Multi-Tenant Isolation across Files & DB
- Asynchronous Background Google Sheets 6-Tab Exporter
- Live Real-Time Telemetry & Monotonic Phase Timers

500K+ HORIZONTAL SCALE-OUT ARCHITECTURE
- Redis Task Queues with Priority Queuing by Source Domain
- Celery / Arq Distributed Harvester Worker Nodes (1..N)
- Residential & Datacenter Proxy Rotation Pool (BrightData)
- PostgreSQL Connection Pooling via PgBouncer Cluster
- pgvector / Qdrant Approximate Nearest Neighbor Indexing
- Apache Kafka / RabbitMQ Streams for Real-Time Consumers
- Multi-Region S3 Object Storage for Raw Ingest Archival
- Prometheus / OpenTelemetry Distributed Tracing & Metrics

3. Multi-Source Ingestion & Verification Matrix

Vertical	Primary Sources & APIs	Ingestion Strategy	Deterministic Verification & Quality Check
Research Papers	arXiv Atom API (cs.AI, cs.LG, cs.CL)	Async XML Chunk Streaming	Live GitHub REST API stars enrichment; strict null on 404
Startups	Seed Startup Registry (50+ firms)	Curated Canonical Harvester	Legal suffix stripping, RapidFuzz WRatio, zero-fabrication headcount
AI Products	Hugging Face Spaces Hub API	Official Spaces REST Client	Enum mapping (FREE, FREEMIUM, PAID, ENTERPRISE)
24h AI News	GNews, Guardian, DataCube, Noozra	Multi-Adapter REST & RSS	Strict 24h window (now-24h ≤ date ≤ now), URL provenance check
24h AI Jobs	Arbeitnow, RapidAPI Indeed, USAJOBS	Multi-Adapter JSON & RSS	AI domain relevance filter, role classifier, 24h date audit

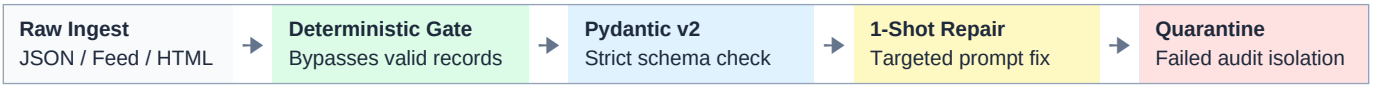
4. Multi-Tier LLM Orchestration & Failover Hierarchy

For unstructured payloads requiring LLM extraction, AETHERIA utilizes a production **LiteLLM Router** configured with automatic exponential backoff, circuit breaking, and sequential tier failover across independent model providers:

Deployment Tier	Model & Provider	Primary Role	Failover Trigger Conditions
Tier 1 (Primary)	Google Gemini 2.0 / 2.5 Flash	High-throughput, structured extraction	HTTP 429 rate limit, 5xx server errors, >4.0s timeout
Tier 2 (Secondary)	Groq / LLaMA 3.1 8B Instant	Ultra-low latency fallback (sub-300ms)	Groq rate limit, connection drop, auth error
Tier 3 (Tertiary)	OpenRouter / GPT-3.5 Turbo	High-reasoning emergency failover	Unrecoverable upstream outage across Tiers 1 & 2

5. Schema Validation, 1-Shot Repair & Diagnostic Quarantine

Structured outputs are validated against strict Pydantic v2 schemas. If validation fails, a targeted 1-shot repair prompt is dispatched with exact validation error diagnostics. If the record remains malformed, it is safely isolated to data/quarantine/{user_id}/{vertical}.jsonl with full error metadata to prevent silent data contamination.



6. Telemetry, Observability Engine & Data Quality Scoring

Every execution records rich, monotonic observability metrics into PostgreSQL and local JSONL logs:
• **pipeline_runs:** Master record tracking run_id, start/finish timestamps, monotonic runtime (ms), harvest counts, structured output, bypass count, LLM token consumption, and failure summaries.
• **pipeline_phase_runs:** High-resolution timers (time.perf_counter()) and record throughput for each phase.
• **pipeline_vertical_runs:** Granular vertical metrics tracking requested vs harvested, fresh vs stale records, and duplicate counts.
• **Data Quality Score Formula (Real-Time %):** Calculated per run via 5 weighted pillars: **Completeness (30%)**, **Schema Validity (25%)**, **Freshness (20%)**, **Duplicate-Free Rate (15%)**, and **Entity Confidence (10%)**.

7. Multi-Tenant User Isolation & Data Partitioning

The platform operates on a **strict multi-tenant architecture**. All filesystem artifacts are partitioned by user UUID: data/{raw,structured,resolved,quarantine,output}/{user_id}/. Scraper deduplication sets are loaded exclusively from the active user's directory, ensuring User A's scrape history never blocks User B from acquiring identical intelligence. Database tables enforce foreign-key user_id scoping with Row-Level Security policies.

Zero-Hallucination Grounding & Validation Policy:
• *"Only extract fields explicitly present in source text. Set missing fields to null. Never fabricate data."*
• **GitHub Star Verification:** Repositories are validated live via GitHub REST API. 404 responses strictly force github_url=null.
• **AI Job Relevance Filter:** Multi-keyword regex and entity checks drop non-AI roles before database insertion.

8. Deterministic Entity Resolution Engine (Phase IV)

Entity names for startups and product creators are canonicalized using a 100% deterministic, zero-LLM matching pipeline. It standardizes legal suffixes (Inc, LLC, PBC, GmbH, Ltd), removes noise, and executes thresholded RapidFuzz WRatio scoring:

Match Score Tier	Resolution Action & Routing	Audit Log Classification
Exact Normalized Match	Immediate auto-merge to canonical entity (bypasses fuzzy engine)	method = seed_match / cross_run_match
WRatio ≥ 90.0	Auto-merge to canonical entity; registers alias for subsequent lookups	method = fuzzy_seed / fuzzy_cross_run
75.0 ≤ WRatio < 90.0	Flagged to manual review log (data/output/{user_id}/manual_review_entities.csv)	method = manual_review (requires_review=true)
WRatio < 75.0	Registered as new canonical entity in run registry	method = new_entity

9. Multi-Layer Deduplication Architecture

Deduplication operates across two coordinated defensive layers:
1. **In-Memory Ingestion Layer:** Unique hashes computed from (company + title + location + source_url) and primary keys (source_name + external_id) prevent intra-run duplicates.
2. **PostgreSQL Consistency Layer:** PostgREST batch upserts target composite unique constraints (user_id, source_url) with resolution=merge-duplicates, providing idempotent writes across parallel runs.

10. Storage Justification & Asynchronous Export Architecture

Primary Storage (Supabase PostgreSQL): Chosen for relational schema enforcement, ACID guarantees, Row-Level Security, and high-efficiency filtering across 11 relational tables.
Asynchronous Google Sheets Export (6 Synchronized Tabs): Google Sheets sync is decoupled from pipeline execution, dispatched in a non-blocking background thread to preserve sub-10s pipeline latency. Accurately provisions 6 worksheet tabs: Startups, Products, Research Papers, Jobs, News, and Entity Mapping Log. Vertical cleanup actions dynamically synchronize with remote Google Sheets tabs, wiping rows while preserving headers.
Why NOT Graph Database (Neo4j) at MVP: Access patterns are vertical-filtered bulk queries and time-windowed freshness lookups. Adding a graph database introduces operational overhead without query advantages at this stage.

11. Pre-Submission Zero-Hallucination Verification Audit

Prior to final submission, src/verify_submission.py performs comprehensive hard-failure assertions: verifying non-null source URLs, validating that papers without GitHub repos strictly have github_url=null, confirming that star counts are verified integers, and enforcing pricing model and job role enum bounds.

Final Architecture Validation Summary:
- **100% Automated Test Suite Passing:** Strict zero-hallucination, schema conformance, and deterministic entity resolution verified.
- **Full Multi-Tenant Data Isolation:** Complete per-user filesystem segregation and database Row-Level Security enforcement.
- **Synchronized Remote Replication:** Real-time Supabase PostgreSQL persistence and background Google Sheets export.